

Name: _____



Finding the n^{th} term

Task A: Finding the n^{th} term of increasing sequences

1) Find the n^{th} term of the sequences below:

a) $10, 17, 24, 31, 38, \dots$

b) $1, 8, 15, 22, 29 \dots$

c) $-10, -7, -4, -1, 2 \dots$

d) $0.9, 1.3, 1.7, 2.1, 2.5 \dots$

e) $\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \dots$

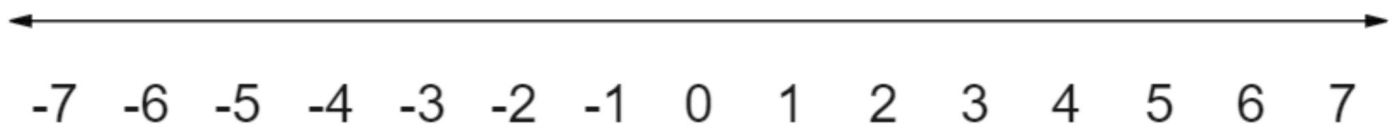
2) Andeep says:

The sequence $-5, -3, -1, 1, 3, \dots$ is $2n - 5$
because the common difference is $+2$ and it
starts at -5



Give two explanations as to why Andeep is wrong.

The number line may help.



Task B: Finding the n^{th} term of decreasing sequences1) Match the sequences to their n^{th} term.

$29 - 5n$

$21, 18, 15, 12, 9, \dots$

$24 - 3n$

$-3, -8, -13, -18, -23, \dots$

$29 - 3n$

$-5, -8, -11, -14, -17, \dots$

$-3n - 2$

$26, 23, 20, 17, 14, \dots$

$-5n + 2$

$24, 19, 14, 9, 4, \dots$

2) Find the n^{th} term of these sequences.

a) $18, 6, -6, -18, -30 \dots$

d) $12, 9, 6, 3, 0 \dots$

b) $-9, -21, -33, -45, -57 \dots$

e) $1.2, 0.9, 0.6, 0.3, 0 \dots$

c) $-15, -18, -21, -24, -27 \dots$

f) $3\frac{1}{2}, 3, 2\frac{1}{2}, 2, 1\frac{1}{2}, \dots$

3) Two students are discussing arithmetic sequences.



Aisha

If you give me any two terms of an arithmetic sequence I can tell you the n^{th} term.

You're right Aisha! If we had 30 and 25 we'd know it's $30, 25, 20, 15, 10, \dots$ with an n^{th} term of $35 - 5n$



Sofia

Write a sentence to explain if Aisha and Sofia are right.

Task C: Finding and using the n^{th} term

1) Find the n^{th} term, then use approximation and counting on to find the term 705 in the arithmetic sequence starting $-25, -15, -5, 5, 15, \dots$

2) Find the n^{th} term, then use approximation and counting on to find the term -554 in the arithmetic sequence starting $-2, -14, -26, -38, -50, \dots$

3) Jacob wants to know the last term positive term in the sequence $36.95 - 0.06n$
Use trial and improvement on your calculator to help him.

